12. State and prove Bernoulli's theorem for a flow of incompressible, non-viscous, and streamlined flow of fluid.
13. Describe the construction and working of venturimeter and obtain an equation for the volume of liquid flowing per second through a wider entry of the tube.

IV. Exercises

1. A capillary of diameter d*mm* is dipped in water such that the water rises to a height of 30*mm*. If the radius of the capillary is made $\left(\frac{2}{3}\right)$ of its previous value, then compute the height up to which water will rise in the new capillary?

   (Answer: 45 mm)

2. A cylinder of length 1.5 m and diameter 4 cm is fixed at one end. A tangential force of 4×10^5 N is applied at the other end. If the rigidity modulus of the cylinder is 6×10^{10} N m^{-2} then, calculate the twist produced in the cylinder. (Answer: 45.60)

3. A spherical soap bubble A of radius 2 cm is formed inside another bubble B of radius 4 cm. Show that the radius of a single soap bubble which maintains the same pressure difference as inside the smaller and outside the larger soap bubble is lesser than radius of both soap bubbles A and B.

4. A block of Ag of mass x kg hanging from a string is immersed in a liquid of relative density 0.72. If the relative density of Ag is 10 and tension in the string is 37.12 N then compute the mass of Ag block. (Answer: $x = 4$ kg)

5. The reading of pressure meter attached with a closed pipe is 5×10^5 N m^{-2}. On opening the valve of the pipe, the reading of the pressure meter is 4.5×10^5 Nm^{-2}. Calculate the speed of the water flowing in the pipe.

   (Answer: 10 ms^{-1})

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